

SUMMARY (narrative by template; all figures from the deterministic engine)

The request for Rs 90,000 over 36 months is referred for manual review under the default policy (v1.0).

Verified monthly income is Rs 39,150 against existing obligations of Rs 2,500; the proposed EMI of Rs 3,076 gives a FOIR of 14.2% (limit 50.0%) and an LTV of 75.0% (limit 80.0%). Risk band: LOW. Reason codes: MISSING_DOCUMENT.

For the borrower:

Ravi Kumar, your application for Rs 90,000 needs a manual check before a decision. Some details in your documents did not fully agree, so an underwriter will review them with you.

Key drivers:

- FOIR 14.2% vs limit 50.0%
- LTV 75.0% vs limit 80.0%
- Verified income Rs 39,150 (volatility 4.1%)
- MISSING_DOCUMENT: Required document missing: dealer_invoice

Next steps:

- Underwriter to resolve the reconciliation flags with the borrower.
- Re-run the workflow once corrected documents are uploaded.

1. DECISION

Decision: REFERRED Confidence: 0.65

Reason codes: MISSING_DOCUMENT

- [warning] MISSING_DOCUMENT: Required document missing: dealer_invoice

1b. WHY THIS DECISION (policy margins + SHAP; words by template)

Ravi Kumar's loan is referred to a human underwriter because Required document missing: dealer_invoice. No hard limit was breached, so an underwriter can still approve it after resolving the flags.

The model's risk band is LOW (score 0.2611 on a 0-1 scale where 0 is safest; a typical synthetic borrower scores 0.437336).

What pushes risk up: ltv (loan / asset value) (75.0% vs a typical 68.1%, +0.012); repayment history score (70.0% vs a typical 71.7%, +0.003).

What pulls risk down: foir (obligations / income) (14.2% vs a typical 41.1%, -0.081); emi-to-income ratio (7.9% vs a typical 34.4%, -0.040); employment stability (95.9% vs a typical 71.8%, -0.024).

The risk score is a secondary signal; the policy rules above decide the outcome.

Mathematics:

PASS FOIR 14.2% vs limit 50.0% (-35.8 pp under)

PASS LTV 75.0% vs limit 80.0% (-5.0 pp under)

PASS age 29 vs minimum 21 (+8 years)

PASS verified monthly income Rs 39,150

risk score 0.261123 = base 0.437336 + sum SHAP -0.176213 (method additive_exact, additivity error 0.0)

$\phi(\text{foir}) = -0.0806$ [value 14.2% vs baseline 41.1%]

$\phi(\text{emi_to_income}) = -0.0398$ [value 7.9% vs baseline 34.4%]

$\phi(\text{employment_stability}) = -0.0241$ [value 95.9% vs baseline 71.8%]

$\phi(\text{income_volatility}) = -0.0183$ [value 4.1% vs baseline 22.4%]

$\phi(\text{income_mismatch_percentage}) = -0.0124$ [value 0.3% vs baseline 12.7%]

2. BORROWER SUMMARY (declared facts)

ref_id: REF-APP-MISSING-007

full_name: Ravi Kumar

age: 29

location_tier: tier_2

employment_type: gig_platform

declared_monthly_income: 39,000.00

loan_request.amount: 90,000.00

loan_request.down_payment: 30,000.00

loan_request.tenure_months: 36

loan_request.vehicle_price: 120,000.00

3. EXTRACTED FACTS (evidence with citations)

full_name = Ravi Kumar conf 0.95 [application_form p.None] auto
monthly_income = 39,000.00 conf 0.95 [application_form p.None] auto
loan_amount = 90,000.00 conf 0.95 [application_form p.None] auto
vehicle_price = 120,000.00 conf 0.95 [application_form p.None] auto
full_name = Ravi Kumar conf 0.85 [kyc.txt p.1] auto
identity = XXXX-XXXX-4321 conf 0.9 [kyc.txt p.1] auto
address = 14 MG Road, Pune 411001 conf 0.85 [kyc.txt p.1] auto
monthly_income = 38,900.00 conf 0.92 [bank_statement.txt p.1] auto
bank_credits = [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0] conf 0.9 [bank_statement.txt p.1] auto
existing_monthly_obligations = 2,500.00 conf 0.92 [bank_statement.txt p.1] auto
address = 14 MG Road, Pune 411001 conf 0.85 [bank_statement.txt p.1] auto
platform_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto
monthly_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto

4. CALCULATED VALUES (deterministic engine, formula v1)

verified_income: 39,150.00
existing_obligations: 2,500.00
emi: 3,075.99
foir: 0.1424
ltv: 0.75
income_volatility: 0.0408
income_stability: 0.9592
loan_to_income: 0.1916
emi_to_income: 0.0786
input.annual_rate: 14.00
input.existing_monthly_obligations: 2,500.00
input.loan_amount: 90,000.00
input.monthly_incomes: [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0]
input.tenure_months: 36
input.vehicle_price: 120,000.00

5. RISK SIGNAL (secondary, never decides)

score 0.2611 band LOW model xgb-v1/fallback

6. AI REASONING

strengths: FOIR 14% is comfortably serviceable
strengths: LTV 75% leaves borrower equity cushion
risks: INCOME_MISMATCH: Monthly income differs across documents by 0.3%
risks: MISSING_DOCUMENT: Required document missing: dealer_invoice
reasoning: Deterministic FOIR 14.24%, LTV 75.00%; XGBoost band LOW (score 0.2611).

7. POLICY

profile default version 1.0
limit.foir_max: 0.5
limit.ltv_max: 0.8
limit.max_tenure_months: 60
limit.min_age: 21

8. HUMAN OVERRIDES

none

SYNTHETIC DATA - demo only. Financial numbers computed deterministically; LLM never sets numbers.